Two Spring Damper System

P12. Numerical Optimization

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1. Problem

Consider the surface $z = (x(x-y))^2 - e^{-x^2-y^2} - \cos(x)$

Consider the plane $x + y + \frac{z}{3} = 0$

Find minimum of z that is on the surface, on the plane, and inside the unit circle. Also find the points x and y at which this minima occurs.

2. Plot of Surfaces

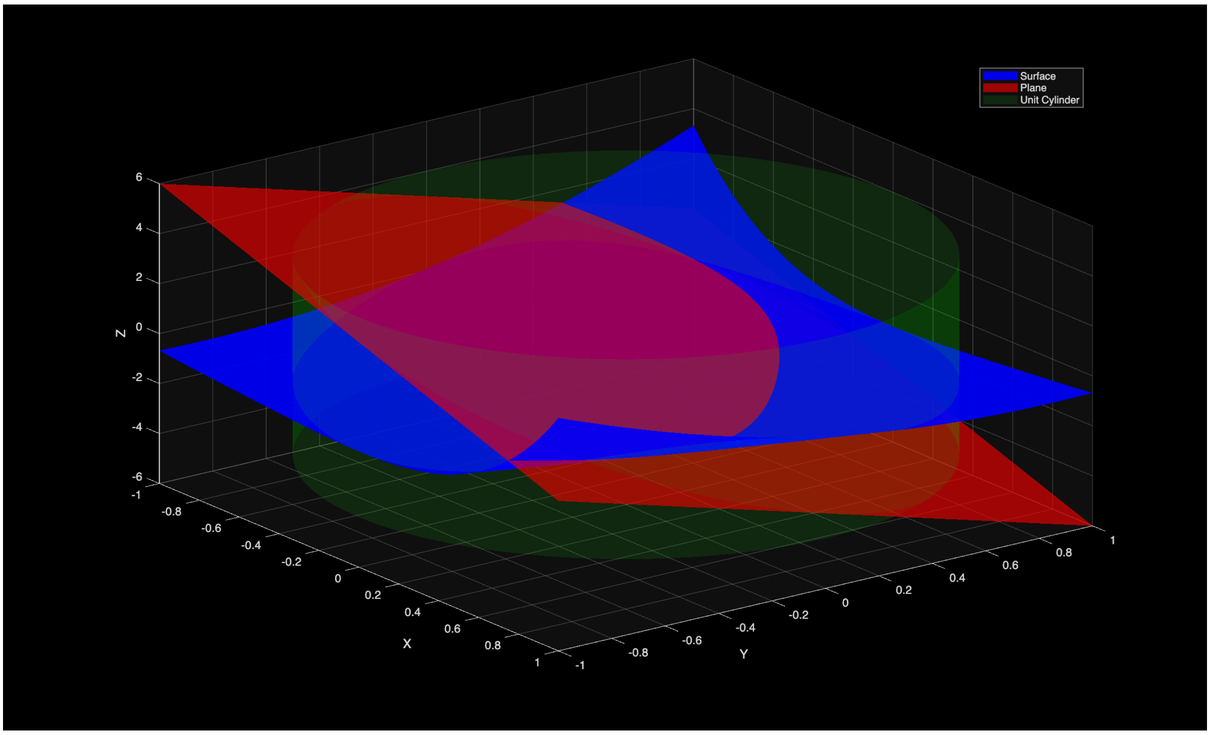

Figure 1: Surfaces

3. GridSearch

```matlab
  function [X_minz, Y_minz, minZ] = gridSearch(X, Y, zSurface, zPlane,
tol)
    % Matlab doesn't support default argumnets so this is the stupid
syntax
    if nargin < 5
        tol = 1e-4;
    end

    [X_minz, Y_minz, minZ] = deal(inf);
    for i = 1:size(X, 1)
        for j = 1:size(X, 2)
            x_ij = X(i, j);
            y_ij = Y(i, j);
            z_ij_surf = zSurface(i, j);
            z_ij_plane = zPlane(i, j);


            if (abs(z_ij_plane - z_ij_surf) >= tol)
                continue
            end

            if (x_ij^2 + y_ij^2 > 1)
                continue
            end

            if (z_ij_plane >= minZ)
                continue
            end

            X_minz = x_ij;
            Y_minz = y_ij;
            minZ = z_ij_plane;
        end
    end
end
```

Listing 1: Grid Search Function Implementation

As I am just comparing the points with some tolerance in this function, we would get better accuracy as we sample more points and change the `numPoints` parameter.

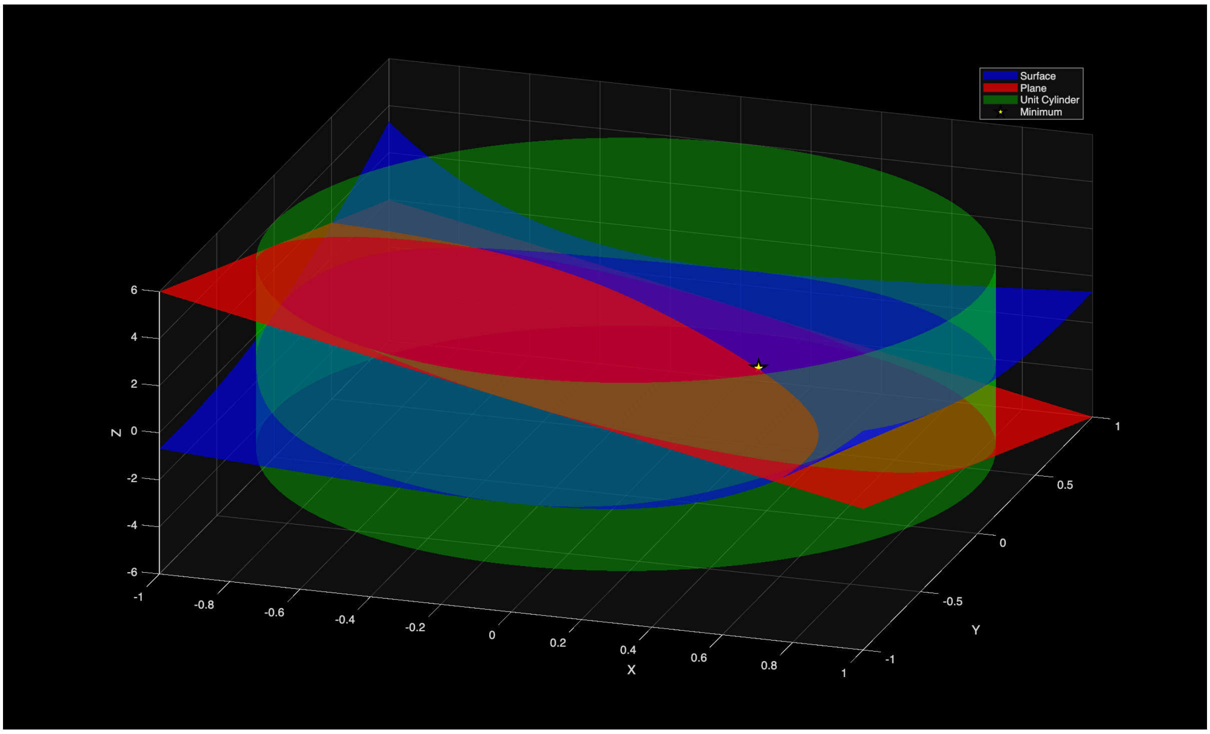

Figure 2: Minima Found Using GridSearch

In the above figure the output was: `Gridsearch Minima(Z): X: 0.28529 Y: 0.31331 Z: -1.7958`

4. `fmincon` Solution

```
function [X_fmin, Y_fmin, Z_fmin] = fminconMinimize()
    obj = @(v) myPlane(v(1), v(2));
    nonlcon = @(v) deal(
        [ v(1)^2 + v(2)^2 - 1 ], % nl ineq
        [ mySurface(v(1), v(2)) - myPlane(v(1), v(2)) ] % nl eq
    );

    v0 = [0, 0];

    [v_opt, z_opt] = fmincon(obj, v0, [], [], [], [], [], [], nonlcon);

    X_fmin = v_opt(1);
    Y_fmin = v_opt(2);
    Z_fmin = z_opt;
end
```

Listing 2: Implementation of `fmincon` Constraints

Output of fmincon function: `fmincon Minima(Z): X: 0.23576 Y: 0.36419 Z: -1.7999`

5. Grid Search Method Error

Two errors were computed for the grid search method:

1. Absolute z error: `abs(Z_fmincon - Z_gridsearch)`
2. Distance Error: `sqrt( (x_fmin - x_gs)^2 + (y_fmin - y_gs)^2 )`

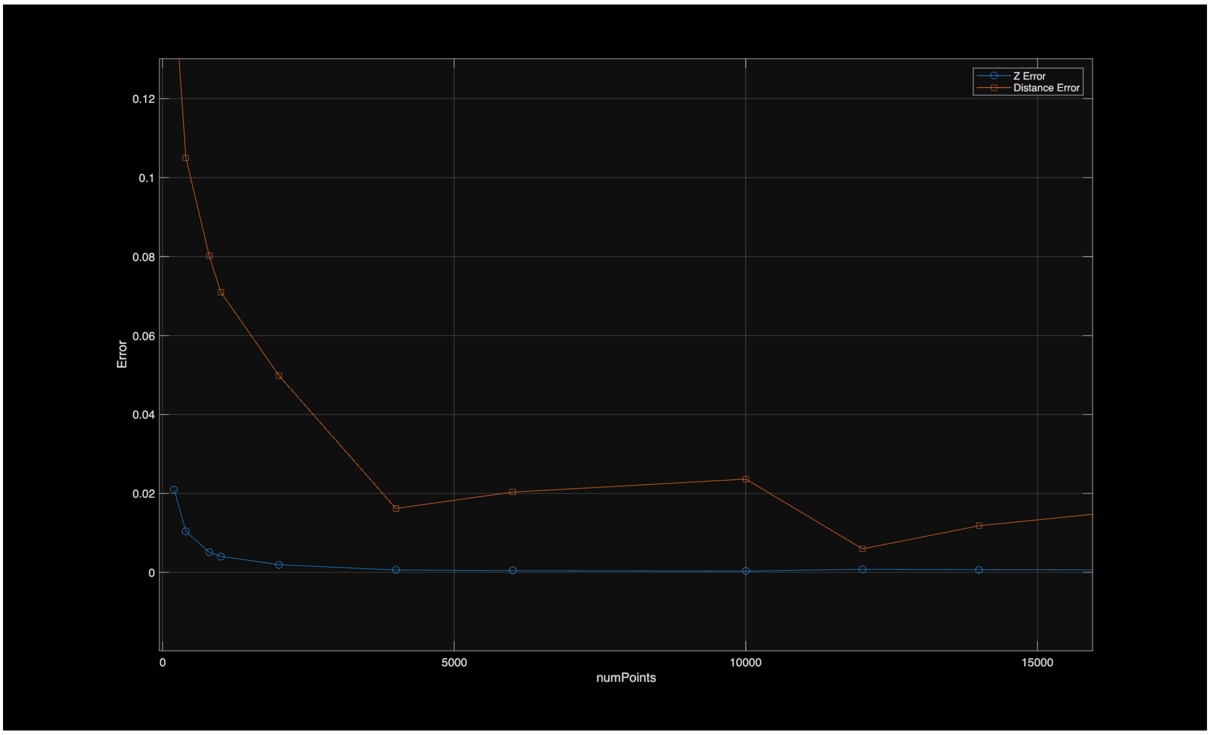

Figure 3: Error in GridSearch